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Engineering Mathematics

13Week [27 November - 03 December]

Quiz 13

Started on	2023-12-04 03:17
State	Finished
Completed on	2023-12-04 09:40
Time taken	6 hours 22 mins
Grade	4.00 out of 5.00 (80%)

Question 1

Correct

Mark 1.00 out of 1.00

Among the answers below choose the one with eigenvalue r and

corresponding eigenvector $\mathbf{u}$ of

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Hint: You do not have to find all eigenvalues and eigenvectors of the matrix.

Select one:

☐ $r = 2, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

☒ $r = 2, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ✓

☐ $r = -1, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

☐ $r = -1, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Your answer is correct.

The correct answer is:

$$r = 2, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Question 2

Correct

Mark 1.00 out of 1.00

Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 5 \\ 1 & -3 \end{bmatrix}.$$

From the answers below first choose the eigenvalues and then corresponding eigenvectors in the same order.

Select one or more:

- ☒ $r_1 = 2$ and $r_2 = -4$ ✓
- ☐ $r_1 = 1$ and $r_2 = -1$
- ☐ $r_1 = -2$ and $r_2 = -1$
- ☐ $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- ☐ $\mathbf{u}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- ☒ $\mathbf{u}_1 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ✓

Your answer is correct.

The correct answers are: $r_1 = 2$ and $r_2 = -4$.

$$\mathbf{u}_1 = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \text{ and } \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Question 3

Correct

Mark 1.00 out of 1.00

Solve the following initial value problem

$$x'(t) = Ax(t), \quad x(0) = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

if the eigenvalues of the matrix A are 1 and -2 and thecorresponding eigenvectors are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Select one:

☐ $x(t) = \begin{bmatrix} e^t + e^{-2t} \\ -e^t - 3e^{-2t} \end{bmatrix}$

☐ $x(t) = \begin{bmatrix} -e^t + 2e^{-2t} \\ 2e^t - 4e^{-2t} \end{bmatrix}$

☒ $x(t) = \begin{bmatrix} -2e^t + 4e^{-2t} \\ -2e^t - 2e^{-2t} \end{bmatrix}$ ✓

☐ $x(t) = \begin{bmatrix} 2e^t \\ -4e^{-2t} \end{bmatrix}$

Your answer is correct.

The correct answer is: $x(t) = \begin{bmatrix} -2e^t + 4e^{-2t} \\ -2e^t - 2e^{-2t} \end{bmatrix}$

Question 4

Incorrect

Mark 0.00 out of 1.00

Two repeated eigenvalue of the matrix

$$A = \begin{bmatrix} 5 & -4 & 4 \\ 12 & -11 & 12 \\ 4 & -4 & 5 \end{bmatrix}$$

are equal to . Among the vectors below find two corresponding linearly independent eigenvectors.

Select one:

- ☐ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
- ☒ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$ ✖
- ☐ $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

Your answer is incorrect.

The correct answer is: $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

Question 5

Correct

Mark 1.00 out of 1.00

Find two linearly independent solutions of the system

$$x'(t) = Ax(t),$$

if the eigenvalues of the matrix A are $2i$ and $-2i$ and the

eigenvector corresponding to the first eigenvalue is $\begin{bmatrix} 1 \\ -1 - i \end{bmatrix}$.

Select one:

☐ $x_1(t) = \begin{bmatrix} e^{2t} \\ -e^{2t} + e^{-2t} \end{bmatrix}$ and

$$x_2(t) = \begin{bmatrix} e^{2t} \\ -e^{2t} - e^{-2t} \end{bmatrix}$$

☐ $x_1(t) = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix}$ and $x_2(t) = \begin{bmatrix} e^{2t} \\ -2e^{2t} \end{bmatrix}$

☐ $x_1(t) = \begin{bmatrix} \cos 2t + \sin 2t \\ -\sin 2t \end{bmatrix}$ and

$$x_2(t) = \begin{bmatrix} \sin 2t - \cos 2t \\ \cos 2t \end{bmatrix}$$

☒ $x_1(t) = \begin{bmatrix} \cos 2t \\ -\cos 2t + \sin 2t \end{bmatrix}$ and

$$x_2(t) = \begin{bmatrix} \sin 2t \\ -\sin 2t - \cos 2t \end{bmatrix} \quad \checkmark$$

Your answer is correct.

The correct answer is: $x_1(t) = \begin{bmatrix} \cos 2t \\ -\cos 2t + \sin 2t \end{bmatrix}$ and

$$x_2(t) = \begin{bmatrix} \sin 2t \\ -\sin 2t - \cos 2t \end{bmatrix}$$